

Dempster Shafer Theory

It is an approach of combining evidence, means for combining degrees of belief ~~de~~ derived from independent items of evidence.

Each fact is having a degree of support between 0 and 1

- 0 no support for the fact
- 1 full support for the fact

Differs from Bayesian approach $\rightarrow \frac{1 - p(x) = \neg p(x)}{p(x) + \neg p(x)} = 1$

- belief in a fact and its negation need not sum to 1
- both values can be 0 — no evidence for/against the fact

Bayesian network works on probability, whereas
Dempster Shafer theory works on plausibility
→ Every proposition is assigned with an interval
- ~~[belief, plausibility]~~

Plausibility (PL) $PL(s) = 1 - Bel(\neg s)$ $\begin{matrix} Bel \\ \hookrightarrow Belief \end{matrix}$

Hence plausibility also ranges in between 0 to 1 and measures the extent to which evidence in favours of $\neg s$

Ex in simple diagnosis problem (—) might consist of {All, Flu, Cold, Pneu}

goal is to add some measure of belief to the elements of (—)

Not all elements are directly supportive of individual elements

If a person is having high fever then all four elements will not play the same role, like all. ~~may~~ may not have huge role

Another key function is used, that is called probability density function (m)

$\{\Theta\} = 1.0$ — all symptom — creating whole room

If correct diagnosis level set to $\{\text{Flu, Cold, Pneum}\}$ then 'Fever' is a piece of evidence

Hence update m as follows

$\{\text{Flu, Cold, Pneum}\} = 0.6$ — impacting half fever
 $\Theta = 0.4$ — all is not used

m_2 ^{is} belief function observing an evidence

'runny nose') Symf. to have runny nose
 $\{\text{All, Flu, Cold}\} = 0.8$ — from entire room of $\Theta = 1$

Remaining room where measure of belief can be considered

$\Theta = 0.2$

Room = Belief + Dis belief

if Belief > Dis belief $\rightarrow$ cause of symptoms can be identified

$$\Theta = \{\theta_1, \theta_2, \dots, \theta_n\}$$

Bayesian concerned with evidence that supports
single conclusion

$$P(\theta_i | E) \quad \text{— judgement-based on probability}$$

DST concerned with evidences which supports
subsets of outcomes

$$\theta_1 \vee \theta_2 \vee \theta_3 \text{ or } \{\theta_1, \theta_2, \theta_3\}$$

— a belief range is given